Payvyn

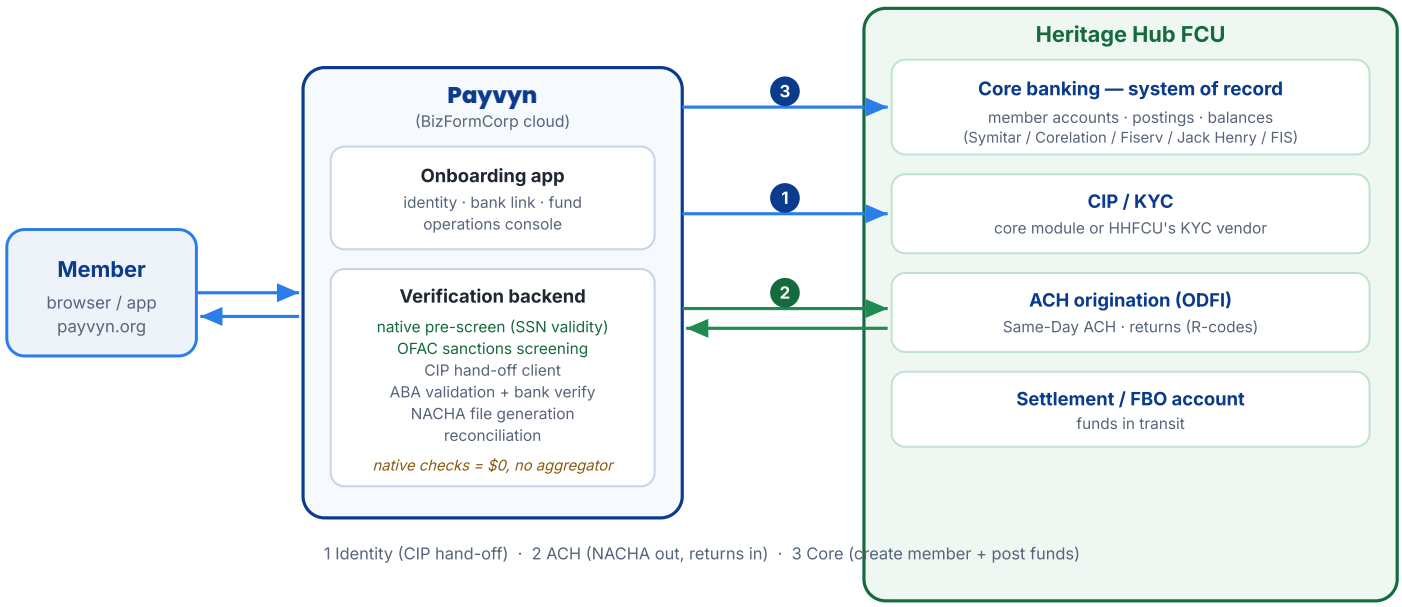
Technical Architecture & Core Access

What Payvyn needs from the HHFCU core, and the end-to-end flow

Payvyn is a member-facing onboarding & verification layer; HHFCU's core is the system of record. Payvyn never replaces or sits inside the core — it **feeds verified members and funded transactions into it and reads back status**, through three well-defined integration points: **CIP/KYC**, **ACH**, and the **core** (account creation + posting). This document lists exactly what is exchanged, the data flow, and a least-privilege way to start.

Payvyn does NOT get access to your core.
In the recommended model, Payvyn **never connects to, signs into, reads from, or writes to your core banking system** — **HHFCU remains the only party that touches the core.** Payvyn hands over verified members and funded-transaction files; your team posts them into the core. The core, member data, and funds stay entirely behind HHFCU's walls. (The optional later API phase uses only a **narrow, scoped client limited to the named operations below** — never open, admin, or standing access, and always subject to HHFCU's approval.)

1 Architecture at a glance



Three integration points. Payvyn runs the free native checks first; only verified members and funded transactions reach the core.

2 End-to-end flow

1. **Onboard** — the member completes Payvyn's form: identity (name, DOB, SSN, address, ID + selfie), links their external U.S. bank, enters the amount to fund.
2. **Native pre-screen** — Payvyn validates the SSN/ITIN and screens the name against the live OFAC lists. Instant, free, before anything reaches HHFCU. Fails stop here.
3. **Identity (point 1)** — Payvyn hands the pre-screened applicant to **HHFCU's CIP/KYC**; HHFCU returns pass / review / fail.
4. **Open account (point 3)** — on pass, a new member/account is created in the core — via a core API, or via a new-account file HHFCU's ops processes.
5. **Bank ownership (point 2)** — Payvyn verifies the external account: two micro-deposits originated over HHFCU's ACH (or instant, if HHFCU offers it).

6. **Fund (point 2)** — Payvyn generates a NACHA file; HHFCU originates an ACH debit from the member's external bank; funds settle to the **FBO/settlement account**.
7. **Post (point 3)** — the settled amount is credited to the member's new core account — via a core posting API, or a settlement/posting file HHFCU posts.
8. **Reconcile** — Payvyn matches settlement + ACH returns (R-codes) against the core and flags exceptions in the operations console.

3 What's exchanged — and what stays inside the core

Access	Purpose	Direction	Data	Protocol options
CIP / KYC submission + result	Authoritative SSN-to-person identity match	`out & in`	name, DOB, SSN, address → decision + reason codes	REST API or SFTP
ACH origination + returns	Micro-deposits & funding debit; return handling	`out & in`	NACHA file out; R-codes (R03/R04…) in	FedLine / correspondent / core ACH / SFTP
Settlement / FBO account	Hold funds in transit before posting	–	account + routing to settle against	account mapping
Member / account creation	Open the new member in the core	`out (write)`	member CIP fields + product/account type	Core API *or* new-account file
Transaction posting	Credit the funded amount to the member	`out (write)`	post credit to member account	Core API *or* settlement/posting file
Posting / transaction status	Reconciliation & confirmations	`in (read)`	posting confirmations, holds	Core API or report feed
Account inquiry (optional)	Show balance / status to the member	`in (read)`	account status, balance	Core API

Key point: this is a list of **data exchanged**, not core access. In the recommended file-based model, member creation + posting are done **by file** (HHFCU's ops posts), so **Payvyn has no access to the core at all** — it only hands HHFCU verified data. Items 1–3 (CIP, ACH, FBO) are required; items 4–7 are things **HHFCU does** from Payvyn's files, or optionally via a scoped API later.

4 Two ways to connect — start least-privilege

A. File-based — RECOMMENDED START

- **Zero core access.** Payvyn never touches the core — it exchanges secure files over **SFTP**.
- Applicant → CIP; NACHA → ACH; a **settlement/posting file** → HHFCU's ops posts to the core.
- HHFCU is the **only** party that writes to the core.
- Fastest to approve — minimal security review; no core API project.

B. API-integrated — PHASE 2

- Direct core APIs for **real-time** account creation, posting, inquiry.
- Via the core's API layer: **SymXchange** (Symitar), **KeyBridge** (Corelation), **DNA APIs** (Fiserv), **jXchange** (Jack Henry), **Code Connect** (FIS).
- Smoother member experience (instant open + post).
- More access → full security review, scoped API client, network tunnel.

5 Security & least privilege

- **Scoped credentials per integration** — a CIP key, SFTP keys, and (phase 2) a core API client limited to **only the operations above**. No standing admin access.
- **Network controls** — IP allowlisting, mutual TLS, and/or a VPN tunnel into HHFCU's network for any core API calls.
- **Data protection** — SSN/account numbers encrypted in transit and at rest; Payvyn retains the minimum (CIP record + masked account); full numbers stay server-side, never in the browser.
- **Audit** — every submission, origination, and posting logged on both sides; returns and exceptions reconciled daily.
- **Separation** — funds settle to an HHFCU-controlled FBO account; Payvyn never holds member funds.

Two things we need from HHFCU to finalize this: (1) which **core banking platform** HHFCU runs (Symitar/Episys, Corelation KeyStone, Fiserv DNA, Jack Henry, FIS…) — it determines the exact API layer; and (2) whether HHFCU prefers to **start file-based** (recommended) or go **direct API**. Everything else is in the Integration Intake Sheet.